Research Statement

Edward Y. Chang, August 2005

I. Accomplishments

Since joined UC Santa Barbara in 1999, my students and I have made significant contributions on three interdisciplinary research topics: 1) *mathematics of perception*, 2) *unified learning paradigm*, and 3) *sensor-data fusion*.

Mathematics of Perception. To support personalization, a search engine must comprehend users' query concepts (or perception), which are subjective and complicated to model. Traditionally, such query-concept comprehension has been performed through a process called "relevance feedback." Our work formulates relevance feedback as a machine-learning problem used with a small, biased training dataset [21]. Unfortunately, traditional machine learning algorithms cannot effectively learn a target concept when the training dataset is small and biased. My team pioneers query-concept learning as the learning of a binary classifier to separate what a user wants from what she or he does not want, sorted out in a projected space [15,25]. We have developed several algorithms to reduce data dimensions [17,23], to maximize the usefulness of selected training instances [15,17,25], to conduct learning on imbalanced datasets [9,18], to accurately account for perceptual similarity [5,7,10], to conduct indexing and learning in a non-metric, high-dimensional space [4,23,24], and to integrate perceptual features with keywords and contextual information [1,3,8,15,16,19,22]. The technology of *mathematics of perception* encompasses an array of algorithms, and has been licensed by major companies including Ask Jeeves, Corbis, Intellireach, Group Technologies, and Trend Micro, etc. for solving their image annotation, retrieval, and filtering problems.

Unified Learning Paradigm. To conduct data mining on massive data, present solutions are neither effective nor efficient. We have recently proposed a unified learning paradigm (ULP) [6], which is an integrated process of supervised, unsupervised, active, and reinforcement learning for acquiring the known and discovering the unknown. Three characteristics distinguish ULP from a traditional hybrid approach such as semi-supervised learning. First, ULP aims to minimize the human effort in collection of quality labeled data. Second, ULP uses the stability of the membership of unlabeled data, together with active learning, to ensure sufficiency of both labeled and unlabeled data, thus guaranteeing the generalization ability of the learned result. Third, ULP uses reinforcement learning (and some other techniques) to access the convergence of the learning process. Preliminary results show ULP to be very effective [2].

ULP operates on an N×N *similarity matrix* formed by N data instances. This similarity matrix is equivalent to the *gram matrix* generated by *kernel methods*. Kernel methods are a core machine learning technique with strong theoretical foundations and excellent empirical successes. One major shortcoming of kernel methods is the cubic training time required with respect to the number of training instances N. In addition, its classification time is O(N), which can present a problem for real-time applications such as military-target and spam-email classification. We have developed novel algorithms to speed up kernel learning's training time, and fast indexing structures to speed up its classification time. We have successfully demonstrated that by carefully decomposing a kernel matrix using *Kronecker factorization*, we can improve training time from $O(N^3)$ to $O(N^2)$ [12]. Presently, we are working with Intel to parallelize the factorization procedure to achieve training time of O(N log N). In addition, our kernel-space indexing work [11] is the first of its kind by indexing data in the projected kernel space (possibly in infinite dimension) to speed up top-k queries. ULP addresses both the effectiveness and efficiency issues of large-scale data mining, and we expect its impact to be very significant.

Sensor-Data Fusion. Sensor and sensor-network research is expected to have significant impact on a broad range of applications relating to national security, health care, etc. In this work, we particularly focus our effort on large-scale sensor-data management and fusion issues. To effectively manage and fuse distributed sensor data, many recent papers propose ad hoc and application-dependent methods. To seek a unified solution, we recognize that resource management in a sensor network is fundamentally a filtering problem: an effective filtering algorithm should filter out the maximum amount of data as long as the query-precision constraints are met at the server. We introduce our hierarchical, dual Kalman filter

architecture as a general and adaptive solution [13,14,19], and are exploring other stochastic, state-space models for supporting sensor data that exhibit non-linear/non-Gaussian characteristics.

II. Future Plan

With our partners in the industry (IBM, Intel, Microsoft, Proximex, and VIMA) and major US airports, we are presently building large-scale video-sensor networks, addressing several large-scale sensor-network management and event-mining issues. To build such a system end-to-end, we are confronted with research challenges of multiple disciplines, including Databases, Data Mining, Computer Vision, Signal Processing, and Statistics. We will build our future work upon our three successful interdisciplinary thrusts summarized above, and continue tackling research problems that cannot be adequately addressed by single-disciplinary approaches.

III. Representative Publications

[1] Multimodal Metadata Fusion Using Causal Strength, Y. Wu, E. Y. Chang, and B. Tseng, ACM International Conference on Multimedia (MM), November 2005.
[2] Formulating Context-dependent Similarity Function, Gang Wu, Navneet Panda, Edward Y. Chang, ACM International Conference on Multimedia (MM), November 2005.
[3] One, Two Class SVMs for Multi-class Image Annotation, K.-S. Goh and E. Y. Chang, IEEE Transactions on Knowledge and Data Engineering (TKDE), 17(10), October 2005.
[4] An Analysis of Transformation on Non-Positive Semidefinite Similarity Matrix for Kernel Machines, G. Wu, E. Y. Chang, and Z. Zhang, International Conference on Machine Learning (ICML)*, Bonn Germany, August 2005.
[5] Formulating Distance Functions via the Kernel Trick, G. Wu, E. Y. Chang, and N. Panda, ACM International Conference on Knowledge Discovery and Data Mining (KDD), August 2005.
[6] A Unified Machine Learning Framework for Large-Scale Personalized Information Management (**invited paper**), E. Y. Chang, Chu-Hong Hoi, Xinjing Wang, W.-Y. Ma, M. Lyu, The 5th Emerging Information Technology Conference, August 2005.
[7] Semantics and Feature Discovery via Confidence-based Dynamic Ensemble, K.-S. Goh, B. Li, and E. Y. Chang, ACM Transactions on Multimedia, 1(2), pp. 168-189, June, 2005.
[8] Support Vector Machine Concept-Dependent Active Learning for Image Retrieval, E. Y. Chang, S. Tong, K.-S. Goh, and C.-W. Chang, IEEE Transactions on Multimedia (accepted).
[9] KBA: Kernel Boundary Alignment Considering Imbalanced Data Distribution, G. Wu and E. Y. Chang, IEEE Transactions on Knowledge and Data Engineering (TKDE), 17(6), pp.786-695, May 2005.
[10] Enhanced Perceptual Distance Functions and Indexing for Image Replica Recognition, A. Qamra, Y. Meng, and E. Y. Chang, IEEE Trans on Pattern Analysis and Machine Intelligence (PAMI), 27(3), pp.379-391, March 2005.
[11] Exploiting Geometric Property for Support Vector Machine Indexing, N. Panda, and E. Y. Chang, SIAM International Conference on Data Mining (SDM), Newport Beach, April 2005.
[12] Kronecker Factorization for Speeding up Kernel Machines, G. Wu, Z. Zhang, and E. Y. Chang, SIAM International Conf. on Data Mining (SDM), Newport Beach, April 2005.
[13] Anatomy of a Multi-camera Security Surveillance System, L. Jiao, Y. Wu, G. Wu, E. Y. Chang, Y.-F. Wang, ACM Multimedia System Journal Special Issue, 10(2), pp. 144-163, November 2004.
[14] Adaptive Stream Resource Management Using Kalman Filters, Jain, E. Y. Chang, and Y.-F. Wang, ACM International Conference on Management of Data (SIGMOD), pp.11-22, Paris, June 2004.
[15] Concept-dependent Multimodal Active Learning for Image Retrieval, K. Goh, E. Y. Chang, and W.-C. Lai, ACM International Conference on Multimedia (MM), pp.564-571, New York, October 2004.
[16] Optimal Multimodal Fusion for Multimedia Data Analysis, Y. Wu, E. Y. Chang, K. C.-C. Chang, and John R. Smith, ACM International Conf. on Multimedia (MM), pp.572-579, New York, October 2004.
[17] MEGA --- The Maximizing Expected Generalization Algorithm for Learning Complex Query Concepts, E. Chang and B. Li, ACM Trans on Information Systems (TOIS), 21(4), pp.347-382, October 2003.
[18] Adaptive Feature-Space Conformal Transformation for Imbalanced-Data Learning, G. Wu and E. Chang, International Conf. on Machine Learning (ICML), pp.816-823, Washington DC, August 2003.
[19] Confidence-based Dynamic Ensemble for Image Annotation and Semantics Discovery, B. Li, K.-S. Goh, and E. Chang, ACM Conf. on Multimedia (MM), pp. 195-206, Berkeley, November 2003.
[20] Multi-camera Spatio-temporal Fusion and Biased Sequence-data Learning for Security Surveillance, G. Wu, Y. Wu, L. Jiao, Y.-F. Wang, and E. Chang, ACM International Conf. on Multimedia (MM), pp. 528-538, Nov. 2003.
[21] Statistical Models and Algorithms for Image Annotation and Retrieval: The Mathematics of Perception (**invited paper**), E. Y. Chang, A book chapter in Digital Multimedia Perception, 2005.
[22] CBSA: Content-based Soft Annotation for Multimodal Image Retrieval Using Bayes Point Machines, E. Chang, K. Goh, G. Sychay, and G. Wu, IEEE Trans. on Circuits and Systems for Video Technology Special Issue on Conceptual and Dynamical Aspects of Multimedia Content Description 13(1), pp.26-38, Jan. 2003.
[23] DynDex: a Dynamic and Non-metric Space Indexer, K. Goh, B. Li, and E. Chang, ACM International Conf. on Multimedia (MM), pp.466-475, Juan Les Pin, France, December 2002.
[24] Clindex: Approximate Similarity Queries in High-Dimensional Spaces, C. Li, E. Y. Chang, H. Garcia-Molina, and Gio Wiederhold, IEEE Trans on Knowledge and Data Engineering (TKDE), 14(4), pp.792-808, July/August 2002.
[25] Support Vector Machine Active Learning for Image Retrieval, S. Tong and E. Chang, ACM International Conference on Multimedia (MM), pp.107-118, Ottawa, October 2001.